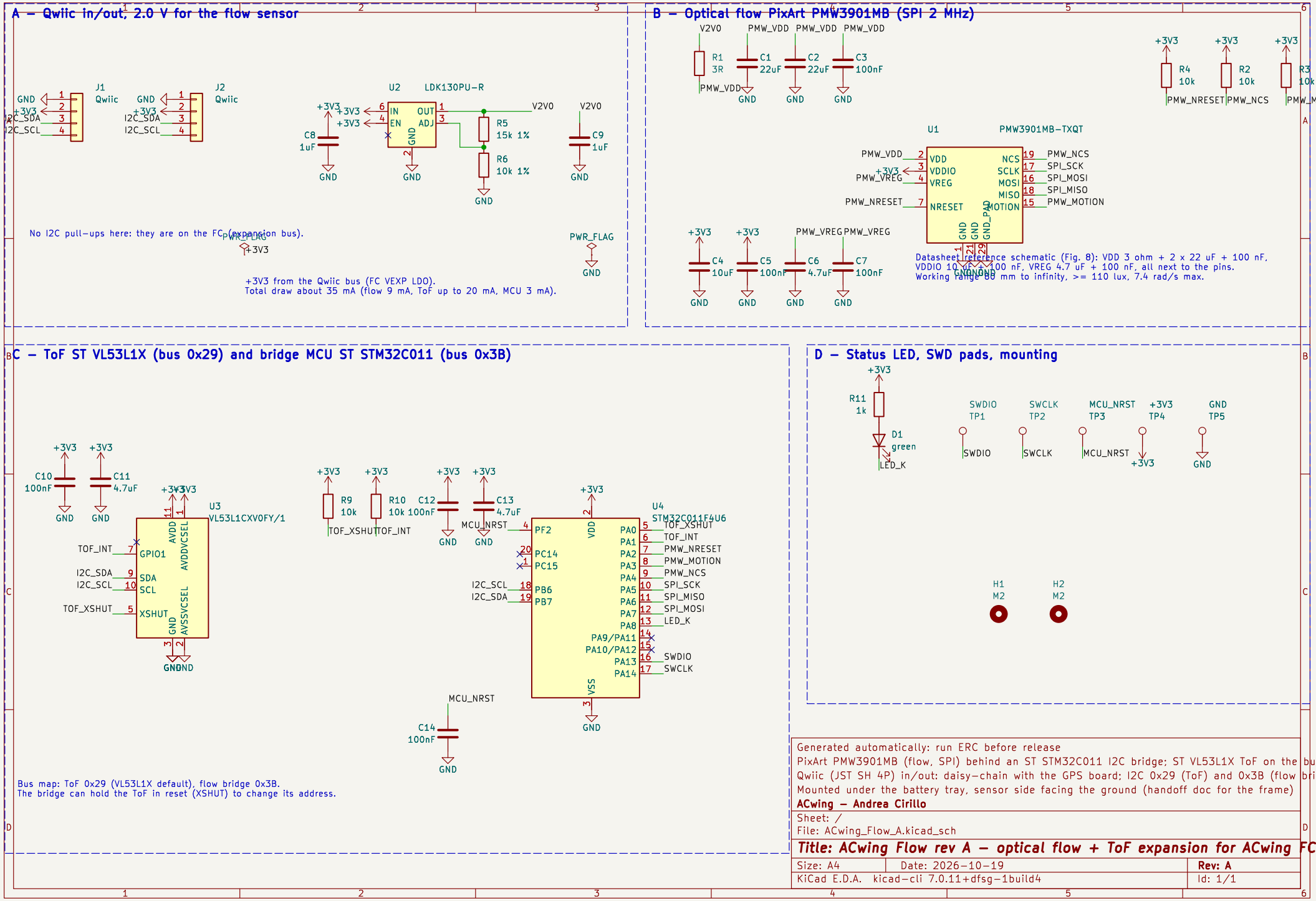

A — Qwiic in/out, 2.0 V for the flow sensor
J1
Qwiic
J2
Qwiic
GND
+3V3
I2C_SDA
I2C_SCL
U2
LDK130PU-R
IN
OUT
EN
ADJ
GND
V2V0
C8
1uF
C9
1uF
R5
15k 1%
R6
10k 1%
No I2C pull-ups here: they are on the FC (expansion bus).
PWR_FLAG
+3V3 from the Qwiic bus (FC VEXP LDO).
Total draw about 35 mA (flow 9 mA, ToF up to 20 mA, MCU 3 mA).
B — Optical flow PixArt PMW3901MB (SPI 2 MHz)
V2V0
PMW_VDD
R1
3R
C1
22uF
C2
22uF
C3
100nF
R4
10k
R2
10k
R3
10k
PMW_NRESET
PMW_NCS
PMW_MOTION
U1
PMW3901MB-TXQT
VDD
VDDIO
VREG
NRESET
NCS
SCLK
MOSI
MISO
MOTION
GND_PAD
PMW_VREG
SPI_SCK
SPI_MOSI
SPI_MISO
C4
10uF
C5
100nF
C6
4.7uF
C7
100nF
Datasheet reference schematic (Fig. 8): VDD 3 ohm + 2 x 22 uF + 100 nF,
VDDIO 10 uF + 100 nF, VREG 4.7 uF + 100 nF, all next to the pins.
Working range 80 mm to infinity, >= 110 lux, 7.4 rad/s max.
C — ToF ST VL53L1X (bus 0x29) and bridge MCU ST STM32C011 (bus 0x3B)
U3
VL53L1CXV0FY/1
AVDD
AVDDVCSEL
AVSSVCSEL
GPIO1
SDA
SCL
XSHUT
TOF_INT
TOF_XSHUT
C10
100nF
C11
4.7uF
R9
10k
R10
10k
C12
100nF
C13
4.7uF
U4
STM32C011F4U6
PF2
PC14
PC15
PB6
PB7
VDD
VSS
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9/PA11
PA10/PA12
PA13
PA14
MCU_NRST
SWDIO
SWCLK
LED_K
C14
100nF
Bus map: ToF 0x29 (VL53L1X default), flow bridge 0x3B.
The bridge can hold the ToF in reset (XSHUT) to change its address.
D — Status LED, SWD pads, mounting
R11
1k
D1
green
TP1
TP2
TP3
TP4
TP5
H1
M2
H2
M2
Generated automatically: run ERC before release
PixArt PMW3901MB (flow, SPI) behind an ST STM32C011 I2C bridge; ST VL53L1X ToF on the bus
Qwiic (JST SH 4P) in/out: daisy-chain with the GPS board; I2C 0x29 (ToF) and 0x3B (flow bridge)
Mounted under the battery tray, sensor side facing the ground (handoff doc for the frame)
ACwing — Andrea Cirillo
Sheet: /
File: ACwing_Flow_A.kicad_sch
Title: ACwing Flow rev A — optical flow + ToF expansion for ACwing FC rev H
Size: A4
Date: 2026-10-19
Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 1/1